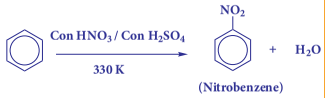
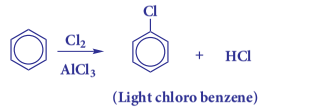
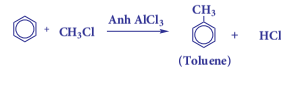
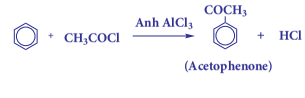
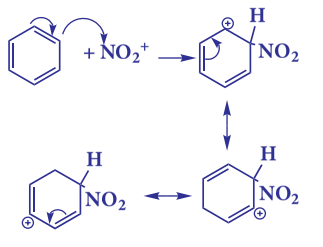
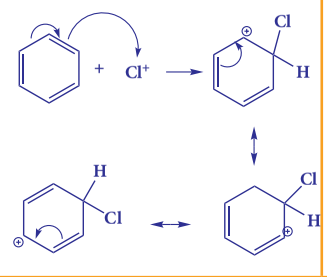
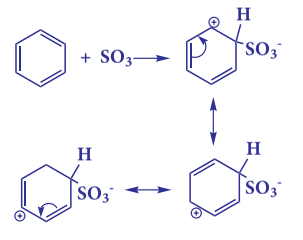
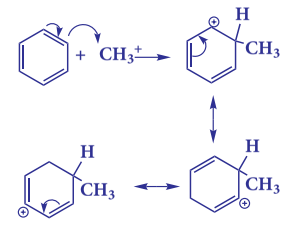
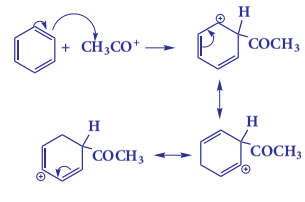
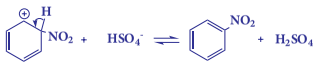
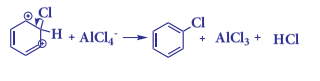
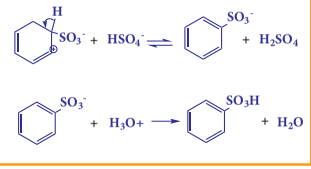
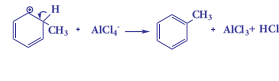
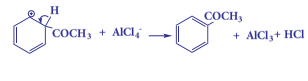
Types of reaction	NITRATION	HALOGENATION	SULPHONATION	FRIEDEL CRAFT'S ALKYLATION	FRIEDEL CRAFT'S ACYLATION
Reagents	Con HNO_3 + Con H_2SO_4	X_2 / Al X_3	Fuming H_2SO_4 $CH_2SO_4 + SO_3$	CH_3Cl + anhydrous $AlCl_3$	CH_3COCl + anhydrous $AlCl_3$
Electro-phile	$-NO_2^+$	$-X^+$ (X = Cl, Br)	$-SO_3$ (Neutral electrophile)	$-\overset{\oplus}{C}H_2$	$-\overset{\oplus}{C}OCH_3$
Over all reaction					
Mechanism Step 1	$NO_3 + H_2SO_4 \rightarrow \overset{\oplus}{N}O_2 + HSO_4 + H_2O$	$AlCl_3 + Cl_2 \rightarrow Cl^+ + \overset{-}{A}lCl_4$	$2H_2SO_4 \rightarrow H_3O^+ + SO_3 + \overset{-}{H}SO_4$	$AlCl_3 + CH_3Cl \rightarrow \overset{+}{C}H_3 + \overset{-}{A}lCl_4$	$AlCl_3 + CH_3COCl \rightarrow \overset{+}{C}H_3CO + \overset{-}{A}lCl_4$
Step 2					
STEP 3					

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